

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MAS-EMPD
Program Name	Master of Engineering Management, Product Design and Development track
Assessment Year	2025
Program Assessment Coordinator	Heng Wang
Year of Current Assessment Plan	2025
No Assessment This Year per Plan	☒
LO(s) Assessed	LO1, LO2, LO3
Presented at program authority meeting date: <i>Please attach minutes</i>	TBD

Methodology

LO#1

- ∄ Learning goal assessed: Graduates will be able to synthesize technical, business, and managerial principles to make strategic decisions throughout the product lifecycle
- ∄ Sample size: All students completing EMGT 470, EMGT 406, BUS 501
- ∄ Semesters in which artifacts were collected: Fall, Spring
- ∄ Artifact: TBD
- ∄ Proficiency ratings
 - ∄ TBD

LO#2

- ∄ Learning goal assessed: Graduates will have the knowledge on relevant methodologies and tools to effectively lead teams and deliver products under constraints
- ∄ Sample size: All students completing a subset of EMGT 575, MMAE 545, MMAE 546, MMAE 547
- ∄ Semesters in which artifacts were collected: Fall, Spring
- ∄ Artifact: For MMAE courses, performance of the students in 4 questions chosen from a pool of questions especially designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ∄ Proficiency ratings
 - For MMAE courses
 - ∄ scores > 75% points indicate **P** proficient.

- ⌵ scores > 55% points indicate **M** meets expectations.
- ⌵ scores < 55% points indicate **D** does not meet expectations.

LO#3

- ⌵ Learning goal assessed: Graduates will be able to formulate ethically and socially responsible strategies that address diverse needs.
- ⌵ Sample size: All students completing a subset of EMGT 547, CHE 541
- ⌵ Semesters in which artifacts were collected: Fall, Spring
- ⌵ Artifact: TBD.
- ⌵ Proficiency ratings
 - ⌵ TBD

Results

Artifact samples

Artifact for LO2 from MMAE 546. Eight questions from a bank of 10 were selected and assigned as a quiz. Sample scores listed below. "C" for correct, "I" for incorrect.

1
Multiple Choice
1 point

Which of the following is a primary advantage of powder metallurgy over traditional casting methods?

- ☒ Ability to produce complex shapes with high dimensional accuracy
- ☐ Lower energy consumption
- ☐ Better surface finish
- ☐ Lower material waste

MS-AM LO1

2
Multiple Choice
1 point

In the context of traditional welding processes, what is the main purpose of flux in arc welding?

- ☒ To protect the weld pool from atmospheric contamination
- ☐ To increase the welding speed
- ☐ To reduce the temperature of the weld pool
- ☐ To increase the strength of the weld

MS-AM LO1

3
Multiple Choice
1 point

In traditional machining processes, which of the following factors most significantly affects tool life?

- ☒ Cutting speed
- ☐ Feed rate
- ☐ Depth of cut
- ☐ Workpiece material hardness

MS-AM LO1

4
Multiple Choice
1 point

What is the primary advantage of using numerical control (NC) in traditional manufacturing processes?

- ☒ Improved consistency and accuracy of parts
- ☐ Reduced need for skilled operators
- ☐ Increased production speed
- ☐ Lower initial equipment costs

MS-AM LO1

1
Multiple Choice
1 point

In the context of Industry 4.0, what is the primary function of the Internet of Things (IoT) in manufacturing?

- ☒ To enable real-time monitoring and control of production processes
- ☐ To replace human workers with robots
- ☐ To eliminate the need for quality control
- ☐ To reduce energy consumption in all manufacturing processes

MS-AM LO2

2
Multiple Choice
1 point

What is a key advantage of using water jet cutting in modern manufacturing?

- ☒ It produces no heat-affected zone in the workpiece
- ☐ It can cut any material regardless of hardness
- ☐ It is faster than laser cutting for all materials
- ☐ It requires no post-processing of cut edges

MS-AM LO2

3
Multiple Choice
1 point

In the context of laser-based manufacturing, what is the primary benefit of using fiber lasers over CO2 lasers?

- ☒ Higher absorption in metals
- ☐ Lower initial equipment cost
- ☐ Larger spot size for wider cuts
- ☐ Better performance on non-metallic materials

MS-AM LO2

4
Multiple Choice
1 point

Which of the following is a key advantage of additive manufacturing (3D printing) over traditional manufacturing methods?

- ☒ Ability to create complex geometries without tooling
- ☐ Higher production speed for large batch sizes
- ☐ Lower energy consumption for all part sizes
- ☐ Better surface finish without post-processing

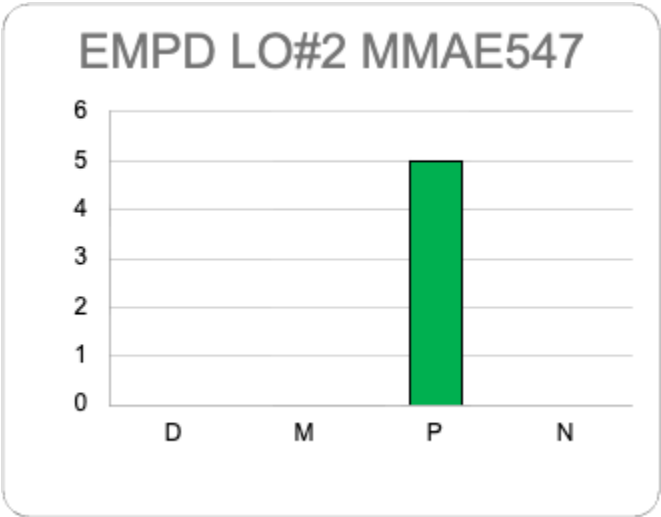
MS-AM LO2

	Sample 1	Sample 2	Sample 3
P	C, I, C, C, C, C, C, C (87.5%)	C, C, C, C, C, C, C, I (87.5%)	
M			

D			
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Data table

This is the first year an assessment plan has been created for this program. EMPD does not have any required courses, so only a total of seven pieces of student artifacts were able to be collected. In addition to two examples included above. The following is the proficiency distribution of for MMAE 547 assessing LO#2, sampled over five students.



Description of findings

Due to the limited sample size, the department will not perform analysis this year.

Continuous Improvement, CI, plans

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☒ After consideration and presentation to the program authority, no CI plans are recommended.

This is the first time MMAE has attempted to do an assessment on this program. MMAE department has realized it must be done after coordinating with other departments. A tentative assessment plan was put together to start the discussion, so a better final plan can be set. MMAE department has started collecting artifacts from its own courses.